



Department of
Administrative Services



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MARCS IT Policy
Radio Patching

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1.0 Introduction

This policy establishes the terms and conditions for patching 2-way radio equipment into the Multi-Agency Radio Communication System (MARCS) infrastructure.

2.0 Purpose

The purpose of this policy is to establish protocols for the use and deployment of radio patching utilizing the MARCS system, ensuring the most effective system usage while maintaining the high grade of service (GOS) necessary for all users of the system.

3.0 Definitions

Radio Patching is one of several terms used to define the connection of an Ohio MARCS radio to another radio or radios, resulting in the ability to establish Interoperable communications between radios and systems not compatible with each other directly.

Some of the other terms used to define this process include “cross-connect”, “interconnect”, or “bridging”.

Of the six levels of Interoperability as originally defined by the Association of Public Safety Communications Officials (APCO), patching is defined as Level Four (4). Patching is also defined in the “Statewide Communications Interoperable Plan” published in April 2021.

Disable is the term used to define a change in status of a radio operating on the MARCS system. Disabling a radio on the MARCS system results in removing the ability of the radio to operate on MARCS.

Gateway Patch allows users to communicate on MARCS from another radio system’s

frequency or talkgroup when needed, without the assistance of a dispatcher or radio operator. The patch is available when needed, while creating no additional usage of airtime resources until the need to communicate arises. There shall be no gateway patches without the expressed written permission of MARCS. This includes LMR to LMR, cellular to LMR, and any other developing technology that enable gateway abilities.

4.0 Background

Ohio MARCS is a performance rated system based on channel and frequency usage at any given time. The performance of MARCS is directly related to how efficiently it is utilized by subscriber agencies. The patching of other radio systems and frequencies to MARCS directly affects the resources available to all users.

Patching a radio system to MARCS is the equivalent of adding the same number of radios in use on that system onto MARCS. In certain situations, this adversely affects the ability of other MARCS users to access the system. Every time a radio is patched to MARCS, even for “monitoring” purposes, a channel may be used at one or more tower sites on the system. This result may minimize the resources available to other agencies on MARCS.

5.0 Administrative Responsibilities

MARCS is responsible for but not limited to the following:

- a) MARCS is responsible for ensuring the necessary network throughput is available at all times for MARCS primary customers.
- b) MARCS provides trunked voice system maintenance and administration services including, but not limited to software or configuration updates and security vulnerability information when received from the manufacturer and a centralized tracking mechanism for addressing software issues associated with the manufacturer.
- c) MARCS performs preventive system maintenance and administration per manufacturer’s recommendations.
- d) MARCS is responsible for ensuring the necessary network throughput is available for only the prescribed or authorized product and affiliated modules.
- e) MARCS is responsible for establishing a process to patch essential radio communication systems through MARCS to optimize emergency communications.
- f) MARCS is responsible for monitoring the use of patches in place and their impact on the MARCS network.
- g) MARCS provides 24/7 Helpdesk support. Call 1-866-OHMARCS (646-2727) or use

the “HELPDESK” talkgroup on MARCS radios.

- h) MARCS provides 24/7 on-call technical staff support, as well as 24/7 call out response when necessary.
- i) MARCS retains the right to shut down a patch at any time.

6.0 Customer Responsibilities

Customers **MUST** be aware of the need for justification of patching and conservation of MARCS system resources.

- a) Agencies employing radio patch capabilities shall advise MARCS of the type of equipment utilized and its intended purpose. This includes defining mobile or fixed capabilities and the systems available to be patched. The responsible agency shall contact the MARCS Operations Center (NOC) Helpdesk via radio – HELPDESK talkgroup or call 1-866-646-2727 if contemplating a standard emergency type patch to MARCS.
- b) Agencies shall be required to provide a single point of contact (Incident Commander or the appointed Communications Unit Leader [COML]). This person shall be in direct control of the patching equipment.
- c) During an incident or exercise, the decision to patch shall be made at the Incident Commander level if possible. An ICS 205 Communications plan shall be submitted to MARCS. The availability of the patch shall be dependent upon MARCS system resources and shall be continually monitored by MARCS.
- d) When the event has terminated, the patch shall be discontinued.

7.0 Cost Assessment

- a) If a long-term or permanent patch is requested, a cost will be assessed to the requesting agency on an annual basis. The cost will be determined using a formula that determines the actual cost incurred by MARCS in the form of towers and talkgroups and/or other resources used by the patching agency.
- b) If the agency terminates the patch prior to a full year’s use, the annual costs will be pro-rated based on the actual period of use.
- c) Long-term or permanent patches will be renewed on an annual basis and new costs may be determined and/or re-assessed based on current rates and/or actual usage at that time.

8.0 Gateway Patch

- a) MARCS recognizes the benefit of gateway patches due to the fact they can be set up without requiring human intervention.
- b) MARCS does not allow gateway patches without prior notification and permission from MARCS. After Gateway patches are approved, the administrator of any gateway patch in place must immediately report the radio ID and location of the patched device. (to include LMR to LMR gateway, cellular to LMR gateway, and any other developing gateway technologies).
- c) MARCS has the right to disable the gateway if it affects system performance.

9.0 Prohibitions

- a) Utilizing MARCS as a means for monitoring a patched radio system from outside the subscriber's daily response region is prohibited.
- b) Utilization of any statewide mass interoperability talkgroups (MCALL/MCOMM, SO-CEN, SO-NE, SO-NW, etc.) for patching is strictly prohibited.
- c) ECOMM talkgroups shall not be used for the purposes of patching.
- d) Utilization of agency specific talkgroups (those used internally by OSHP/ODNR/OEMA tactical or dispatch talkgroup, local police/fire/EMS) is strictly prohibited without express permission from the owning agency. (e.g., If an OSHP dispatcher needs to patch an OSHP dispatch to a county sheriff dispatch, the OSHP dispatcher must first ask the county sheriff dispatcher if it can be done and precisely which talkgroup can be used for the patch.) Exceptions shall be made only if the incident lies within the command jurisdiction of the owning agency. It is up to the talkgroup owning agency to decide which talkgroup to use for the patch. Utilization of the agency specific dispatch is still prohibited.

10.0 Enforcement

- a) Violation of the above rules shall result in a thorough administrative investigation by agents of the MARCS program office.
- b) Violations of this policy shall result in the affected device being disabled by MARCS. Disabling a radio results in the device not being able to transmit on MARCS.
- c) If a patch is detected that is found to be adversely affecting the performance of MARCS, and the means to contact the Communication Unit Leader (COML) fails, then MARCS reserves the right to remotely “inhibit” or turn off the radio that is being used to patch to other systems. This means shall be used as a last resort only when necessary to preserve MARCS system performance for all other users.
- d) If it is determined that an unauthorized patch resulted in a significant amount of depletion of MARCS resources, a cost will also be assessed to the offending agency for the amount of actual cost determined to have been incurred by the MARCS program.
- e) It is not the intention of MARCS to deter the use of radio patching capabilities but rather, maximize usage of the system statewide. MARCS is responsible for ensuring system performance is maximized at all times for all users.

11.0 Effective Date

The current version of this policy was approved and adopted by the MARCS Steering Committee on May 9, 2023.

12.0 Revision History

DATE	Description of Change
03/15/2008	Original document
12/16/2021	Adding revision history fields, updating letterhead, removing 3.5 related information
4/25/2023	Modified definition for Gateway Patch, revised call out availability as to all customers, added potential cost remedy where a prohibited patch is detected, and a potential cost determination and assessment where a long-term or permanent patch is desired

13.0 Attachments

Agency Gateway Information and Agreement form



Agency Gateway Information and Agreement

Date: Click or tap to enter a date.

Requesting Agency:	Enter MARCS Customer Agency Name Here.
Agency Contact:	Enter Agency Contact Information Here Including Contact Information.
MARCS Customer ID:	Enter MARCS Customer ID.
MARCS Device ID:	Enter MARCS Device ID.
Device Model and Serial Number:	Enter Device Model and Serial Number here.
MARCS Talkgroup Requested:	Enter Requested MARCS Talkgroup Information Here.

Solution Design Description: Please describe proposed solution in detail in this area.

Requested Gateway Activation Dates:

From: Click or tap to enter a date. To: Click or tap to enter a date.

Authorized Agency Contact Information:

Name: Enter Authorized Agency Contact Information here.

Title: Enter Title of Authorized Agency Contact here.

Date: Click or tap to enter a date.

This agreement is between the Ohio Department of Administrative Services, Office of Information Technology, Multi-Agency Radio Communication System (MARCS) and the undersigned (SUBSCRIBER). The subscribing agency agrees to the terms and conditions of this agreement as posted on the MARCS Services website (MMP-22) and to the current rates in effect at the time of billing, if applicable. Any person executing this Agreement in a representative capacity hereby states that he/she has been duly authorized by his/her authorizing authority to execute this Agreement on such authorizing authority's behalf.

Signed _____ Date _____
(Print name below signature)